[INCH-POUND]
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COMMERCIAL ITEM DESCRIPTION

SCALING AND CHIPPING TOOL, ROTARY, PORTABLE, ELECTRIC, AND CUTTER BUNDLES

This Commercial Item Description is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

Scope and Classification. This commercial item description covers scaling and chipping tools intended for use in removing rust, scale, paint, and other hard deposits from steel, iron, concrete, and similar surfaces through the action of rotating cutter bundles.

Salient Characteristics.

Tool requirements:

The tool shall be of the manufacturer's current design (see Figure 1).

All wiring, flexible cord, and attachment plug cap shall meet the requirements of Underwriters' Laboratories Standard No. 45. The cord length shall be at least 10 feet long.

The tool shall have basic insulation.

The tool shall have a 2 pole, 3-wire grounding system that meets the requirements of Underwriters' Laboratories Standard No. 498.

The motor shall be of the universal series wound type capable of operating on 120 volts AC/DC. The motor shall develop 3/8 horsepower or more at the cutter bundle shaft with cutter bundles removed.

The tool shall be electromagnetic interference suppressed in accordance with MIL-STD-461, when specified.

The switch shall be double pole and of the quick break type with provisions for locking in the "ON" position.

All rotating parts shall be provided with precision quality ball bearings or ball and roller bearings.

The tool shall be of the double cutter type.

The tool shall have a sturdy supporting wheel, or wheels, mounted on the body of the tool.

The handle shall be of the closed grip type incorporating a trigger-type switch. A second handle shall be mounted on the tool in such a way as to permit the operator to exert maximum downward pressure on the cutter bundles.

Cutter bundles shall be covered by guards such that the operator is protected from the failure of rotating parts. The guard shall be situated in such a manner that the cutter bundles are exposed for 180° $\pm$ 10° around the periphery, and the exposed portion faces the work surface. The guard shall not interfere with the tool's operation or with changing the cutter bundles.

FSC 5130

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The cutter bundle drive spindle shall be 0.5 inch, +0.000, -0.005 inch thick square and shall accommodate 2 cutter bundles.

Cutter bundle rotation shall be parallel to the length of the tool. The fit of the bundles on the drive spindle shall be a sliding fit.

The tool shall be provided with a means of easily regulating the depth of cut. That device shall be wear resistant.

All parts shall be replaceable and adjustable without modification of the parts or the tool.

An identification plate, securely fastened to, or as an integral part of the housing, shall be provided. The plate shall contain the following information: manufacturer's name, trademark, country of origin, model and serial number, type of current, voltage, load current (rated amperes), and non-load rpm of spindle shaft.

The tool shall be so designed and constructed that it is fully protected from the harmful effects of debris created by the tool's operation. All parts requiring protection from dust and debris shall be enclosed.

The tool shall be provided with one pair of rebuildable cutter bundles.

The tool shall be supplied with any tools, equipment, and accessories necessary to assure efficient operation in accordance with the operating instructions and any tools, equipment, and accessories normally supplied commercially. The tool shall be supplied with a complete set of the manufacturer's current operating and maintenance instructions, parts list breakdown, and any other information necessary for the safe and efficient operation of the tool.

The tool shall be supplied with a carrying case capable of securely holding and compactly housing the tool and all accessory components.

Cutter bundle requirements (see Figure 2): The cutter bundle assembly shall be rebuildable and shall be constructed as to permit simple and quick replacement of components. The cutter bundle design shall permit full use of the cutter teeth to their root diameter. All replacement parts shall be constructed so that worn parts may be replaced without modification. The cutter bundle assembly shall consist of a bundle frame, cutters, washers, spacers, cutter shafts, and spring retainers as follows:

Bundle frame: The bundle frame shall consist of two (2) flanges and (1) hub. The flanges shall be welded to the hub. The flanges shall be 3.188 inches $\pm$ 0.016 inch in diameter and 0.125 inch $\pm$ 0.005 inch thick. The overall width of the bundle frame shall be 2.125 inches $\pm$ 0.016 inch. The hub shall be made from square mechanical tubing with a 0.502/0.515 finished square bore. The material for the flanges and hub shall be steel. The finished weldment shall be heat treated and tempered to Rockwell C 59-61. The bundle frame shall be designed so that the bundle assembly may be mounted on either end of the square hub on the scaling and chipping tool by means of a sliding fit.

Cutters: The cutters shall be milled or stamped steel discs with a minimum of 18 teeth equally spaced around the periphery. Diameter of the cutter shall be 1.24 inches $\pm$ 0.080 inch. The thickness shall be 0.125 inch $\pm$ 0.016 inch. The bore shall be 0.625 inch minimum. The cutters shall be hardened to Rockwell C 57 to 60. With cutters mounted, the operational diameter of the cutter bundle assembly shall be 3.875 inches maximum. Each cutter bundle assembly shall consist of six (6) rows of cutters. Each cutter row shall consist of seven (7) cutters alternately spaced with seven (7) spacers.

Washers: One (1) washer per cutter shaft placed at either end of cutter shaft depending on cutter position. The material shall be steel. The outer diameter shall be 0.810 inch $\pm$ 0.020 inch. The inner diameter shall be 0.441 inch $\pm$ 0.003 inch. The thickness shall be .120 inch $\pm$ 0.005 inch.

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Spacers: Six (6) per cutter shaft alternating in relative position with cutters on adjoining axial members. The material shall be steel. The outer diameter shall be 0.843 inch ± 0.005 inch. The inner diameter shall be 0.439 inch ± 0.005 inch. The thickness shall be 0.134 inch ± 0.006 inch.. The spacers shall be case hardened to Rockwell 15N 86-88.

Cutter Shafts: The cutter shaft shall be 0.425 to 0.430 inches in diameter. The cutter shaft material shall be steel hardened to Rockwell C 54-59. Each cutter bundle assembly shall consist of (6) cutter shafts appropriately located and equally spaced around the flanges.

Spring retainers: The spring retainer(s) or snap ring(s) shall be used to secure the cutter shafts. The retainer(s) or snap ring(s) shall not work loose, deform, or break during the operation of the cutter bundle assembly.

Interchangeability: The cutter bundle assembly shall be functionally and mechanically compatible with the Porter Cable Corporation Scaling and Chipping Tool, part number 727 .

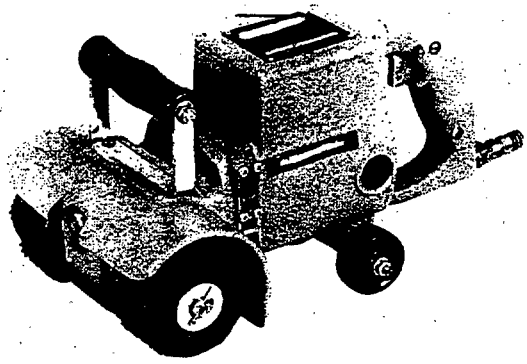


Figure 1. Scaling and Chipping Tool.

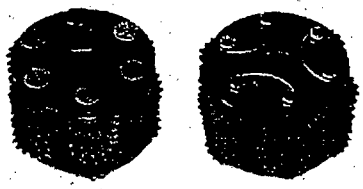


Figure 2. Cutter Bundle Assembly.

Quality assurance provisions: Quality assurance provisions shall consist of:

(1) First article inspection.
(2) Quality conformance inspection.

The first article sample shall be one tool. The first article inspection shall consist of (a) visual and dimensional examinations of the item to verify conformance to the requirements specified herein, and (b) the simulated service test. Failure of the first article to pass the simulation service test shall be cause for rejection.

Quality conformance inspection shall consist of (a) visual and dimensional examinations of the item to verify conformance to the requirements specified herein, and (b) the first five hours of the simulated service test performed on one sample drawn randomly. Quality conformance inspection shall not be required on definite quantity contracts.

Simulated service test: The scaling and chipping tool shall satisfactorily complete a service test of at least 150 hours of actual operation. Non-operational periods may be interspersed with the periods of operation. During operation, the tool or test log material (hot rolled steel plate SAE 1010, 1018, or 1020) shall be oscillated forward and backward in a manner that simulates actual service conditions. The average rate of chip removal during the first five hours of operation shall be 0.5 pound per hour for first five hours. After completion of the service test, the scaling and chipping tool shall be disassembled and examined for evidence of wear of motor bearing(s), shaft(s), insecure fastenings and evidence of lubricant leakage. Excessive wear of motor bearing(s), shaft(s) affecting the life expectancy of the tool, insecure fastenings, or excessive loss of lubricant affecting the life expectancy of the tool shall be cause for rejection. Upon completion of the simulated service test, the cutter bundles shall be disassembled and components examined. Any cutter bundle having broken, cracked, missing, or deformed parts, other than normal wear, shall be rejected. Failure to comply with the examination(s) or the above test shall be cause for rejection.

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Note:

When only cutter bundles are procured under this Commercial Item Description, failure of the tool used to test the cutter bundles shall not be cause for rejection of the cutter bundles unless the failure of the tool was caused by the cutter bundles.

Workmanship: The scaling and chipping tool and accessory components shall be free of any defects which may impair serviceability, durability, or appearance.

Regulatory requirements. In accordance with Section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing suppliers' employees to undue hazards from the recovered materials.

Preservation, packaging, packing, labeling, and marking. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

Notes:

Assigned National Stock Numbers (NSNs)

5130-01-087-6819 - Electromagnetic interference suppression not required.
5130-00-288-6577 - Electromagnetic interference suppression required.
5130-00-287-5199 - One set of two cutter bundles only.

Preparing Activity:

GSA-FSS

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